



Book I. Prop. VIII. Theor.

If two triangles have two sides of the one respectively equal to two sides of the other (— = — and — = —) and also their bases (— = —), equal; then the angles (◼ and ◼) contained by their equal sides are also equal.

If the equal bases — and — be conceived to be placed one upon the other, so that the triangles shall lie at the same side of them, and that the equal sides — and —, — and — be conterminous, the vertex of the one must fall on the vertex of the other; for to suppose them not coincident would contradict the last proposition.

Therefore sides — and —, being coincident with — and —, $\therefore \blacktriangle = \blacktriangle$.

Q. E. D.